

Seminar 2.

Def (lim of seq). Let $\{a_n\}$ be a sequence of real numbers. A number a (L) is the limit of $\{a_n\}$, written $\lim_{n \rightarrow \infty} a_n = a$ if

$$\forall \varepsilon > 0 \quad \exists N = N(\varepsilon) \in \mathbb{N} \quad (\forall n > N : |a_n - a| < \varepsilon)$$

- take $\varepsilon > 0$
- $|a_n - a|$ simplify

- $|a_n - a| < \varepsilon \rightarrow$ solve this inequality with respect to n .
- choose $N \in \mathbb{N} \rightarrow N(\varepsilon)$.

□

Problem 1a)

$$\lim_{n \rightarrow \infty} \frac{3}{n+2} = 0$$

$$a_n = \frac{3}{n+2}$$

$$a = 0$$

Proof:

- $\forall \varepsilon > 0$
- $\left| \frac{3}{n+2} - 0 \right|$
- $\left| \underbrace{\frac{3}{n+2}}_{a_n} - \underbrace{0}_a \right| < \varepsilon$
- $\frac{3}{n+2} < \varepsilon$
- $\frac{n+2}{3} > \frac{1}{\varepsilon} \quad | \cdot 3$

$$n+2 > \frac{3}{\varepsilon}$$

$$n > \frac{3}{\varepsilon} - 2$$

$$n+2 > n$$

$$N = \left\lceil \frac{3}{\varepsilon} - 2 \right\rceil$$

$$N \geq \left\lceil \frac{3}{\varepsilon} \right\rceil$$

$$N = \left\lceil \frac{3-2\varepsilon}{\varepsilon} \right\rceil$$

$$\left| \frac{3}{n+2} - 0 \right| = \frac{3}{n+2} < \underline{\underline{\varepsilon}} < \frac{3}{\varepsilon}$$

□

Verification: Let $\forall n > N \Rightarrow$

$$\Rightarrow n > N \geq \frac{3}{\varepsilon} > 0$$

$$n > \frac{3}{\epsilon}$$

$$\epsilon n > 3 \quad | : n$$

$$\boxed{\epsilon > \frac{3}{n}}$$

$$\left| \frac{3}{n+2} - 0 \right| < \epsilon \iff n \rightarrow \infty$$

Problem 2a, b.

$$a) \lim_{n \rightarrow \infty} \frac{3n+1}{n+2} = 3$$

$$a_n = \frac{3n+1}{n+2}$$

$$a = 3$$

Proof: $\forall \epsilon > 0$

$$\bullet \left| \frac{3n+1}{n+2} - 3 \right|$$

$$\bullet \left| \frac{3n+1-3n-6}{n+2} \right| = \left| \frac{-5}{n+2} \right| = \frac{5}{n+2}$$

$$\bullet \frac{5}{n+2} < \epsilon$$

$$\frac{n+2}{5} > \frac{1}{\epsilon}$$

$$n+2 > \frac{5}{\epsilon}$$

$$N = \left\lceil \frac{5}{\epsilon} - 2 \right\rceil \Rightarrow$$

$$N \geq \left\lceil \frac{5}{\epsilon} \right\rceil$$

$$b) \lim_{n \rightarrow \infty} \frac{5n + (-1)^n}{3n+4} = \frac{5}{3}$$

$$\bullet a_n = \frac{5n + (-1)^n}{3n+4}$$

$$a = 5/3$$

Proof: $\forall \epsilon > 0$

by def $\rightarrow$
very very small
number

$$\bullet \left| \frac{5n + (-1)^n}{3n+4} - \frac{5}{3} \right| =$$

$$= \left| \frac{3(5n + (-1)^n) - 15n - 20}{3(3n+4)} \right| = \left| \frac{3(-1)^n - 20}{3(3n+4)} \right| = \frac{23}{3n+12} \quad \checkmark \checkmark$$

$$\bullet \frac{23}{3n+12} < \epsilon$$

$$n=1$$

$$n=2$$

$$n=3$$

$$g_{n+12} > \overline{\epsilon}$$

$$g_n > \frac{23}{\epsilon} - 12$$

$$n > \frac{23}{9\epsilon} - \frac{12}{9}$$

$$, N = \left\lceil \frac{23}{9\epsilon} - \frac{12}{9} \right\rceil \Rightarrow N \geq \left\lceil \frac{23}{9\epsilon} \right\rceil$$

Problem 3a

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} + 4}{2\sqrt{n} + 6} = \frac{1}{2}$$

Solution:

$$\begin{aligned} & \forall \epsilon > 0 \\ & \left| \frac{\sqrt{n} + 4}{2\sqrt{n} + 6} - \frac{1}{2} \right| < \epsilon \end{aligned}$$

$$\frac{2\sqrt{n} + 8 - 2\sqrt{n} - 6}{4\sqrt{n} + 12} < \epsilon$$

$$\frac{2}{4\sqrt{n} + 12} < \epsilon$$

$$\frac{4\sqrt{n} + 12}{2} > \frac{1}{\epsilon} \cdot 2$$

$$\begin{aligned} 4\sqrt{n} + 12 &> \frac{2}{\epsilon} \\ 4\sqrt{n} &> \frac{2}{\epsilon} - 12 \\ \sqrt{n} &> \frac{1}{2\epsilon} - 3 \end{aligned}$$

$$n > \frac{1}{4\epsilon^2}$$

$$N \geq \left\lceil \frac{1}{4\epsilon^2} \right\rceil$$

Problem 4.

A sequence $\{a_n\}$ is divergent if no real number is its limit, that is if:

$$\forall a \in \mathbb{R} \quad \exists \epsilon > 0 \quad \forall N \in \mathbb{N} \quad \exists n > N : |a_n - a| \geq \epsilon.$$

Solution: suppose, we have a contradiction $\Rightarrow$

$\Rightarrow$ our seq. converges. let $a \in \mathbb{R}$, with

$$\text{that } \lim_{n \rightarrow \infty} a_n = a, \quad a_n = (-1)^n$$

$$\Downarrow$$

$$\lim_{n \rightarrow \infty} (-1)^n = a.$$

$\therefore \nexists$

$\forall \epsilon > 0$

- $a_n = (-1)^n$

- $\forall \epsilon > 0$

$$\underline{\epsilon = 1.}$$

$$|a_n - a| < \epsilon$$

$$|a_n - a| < 1.$$

$$|(-1)^n - a| < 1.$$

$$n_1 = 2N+2 (\text{even})$$

$$n_2 = 2N+3 (\text{odd})$$

$$(-1)^{n_1} = (-1)^{2N+2} = 1.$$

$$(-1)^{n_2} = (-1)^{2N+3} = -1.$$

$$|1 - a| < \epsilon$$

$$|1 - a| < 1$$

✓

$$|-1 - a| < \epsilon$$

$$|-1 - a| < 1$$

✓

contradiction.

$\Rightarrow$ seq. diverges.